



Science

Stage 8

Paper 1

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

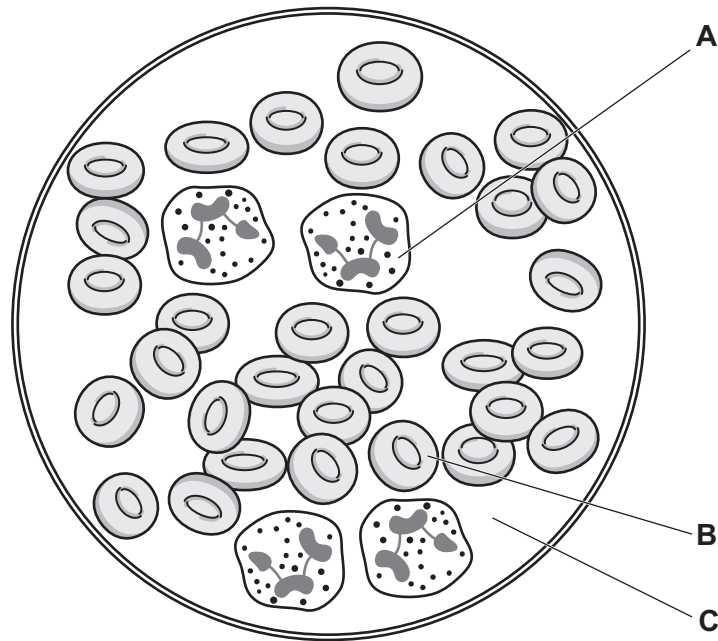
- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Pierre uses a microscope to look at some blood.

He draws an image of what he observes.



- (a) Complete the table about the functions of the components of the blood.

function	letter
transports nutrients	
transports oxygen	
protects against pathogens	

[2]

- (b) Pierre says,

'I predict that animals with a larger mass will have red blood cells with a larger diameter.'

He decides to look for secondary information from the internet.

- (i) Give **one advantage** and **one disadvantage** of using secondary information from the internet.

advantage

.....

disadvantage

.....

[2]

(ii) Here is the information Pierre finds.

animal	average mass of an adult in kg	average diameter of red blood cells in μm
dog	25	7.0
horse	300	5.7
sheep	80	4.5
goat	90	3.2
human	65	7.1
cat	4.5	5.8
rabbit	1.5	6.1

Tick (✓) to show if this information supports Pierre's prediction.

yes ☐ no ☐

Explain your answer.

Use information from the table.

.....

.....

.....

[1]

(c) Complete the sentence.

Choose from the list.

breathing

diffusion

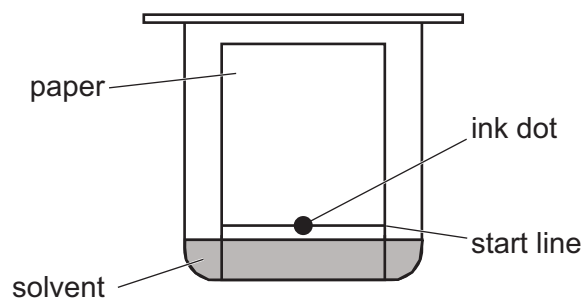
digestion

respiration

Gases move between blood and the air in the lungs by [1]

2 Carlos investigates the colours in ink.

(a) The diagram shows his equipment at the start of the investigation.



(i) Name this method of separating the colours in ink.

..... [1]

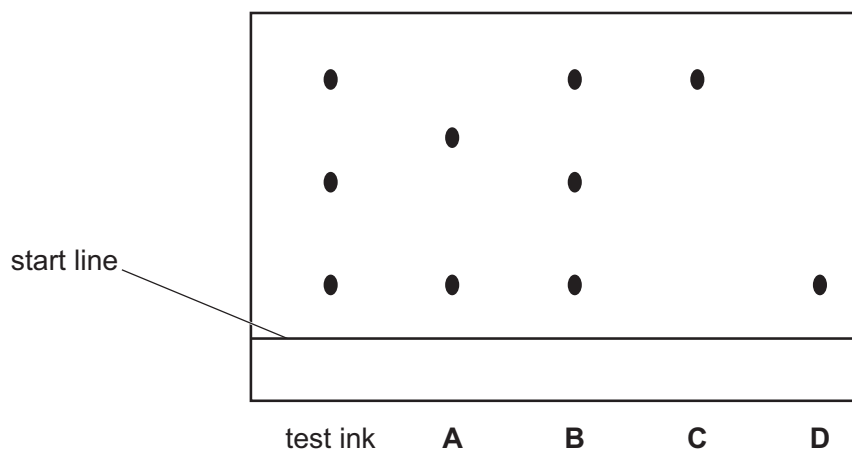
(ii) Explain why the ink dot must be above the solvent.

.....

 [2]

(b) Carlos separates the colours from five different samples of ink.

Look at the diagram that shows the results of these separations.



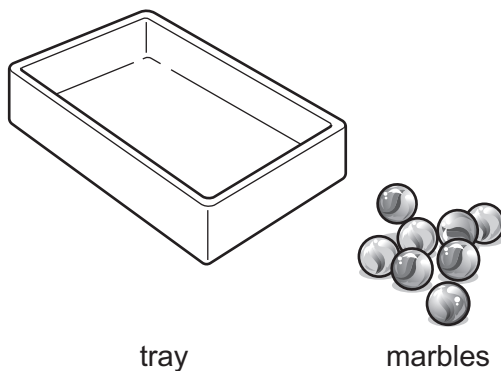
Which ink contains the same colours as the test ink?

Choose from **A**, **B**, **C** or **D**

Explain your answer.

.....
 [2]

- 3 Aiko uses a tray and some marbles to model gas pressure **in a container**.



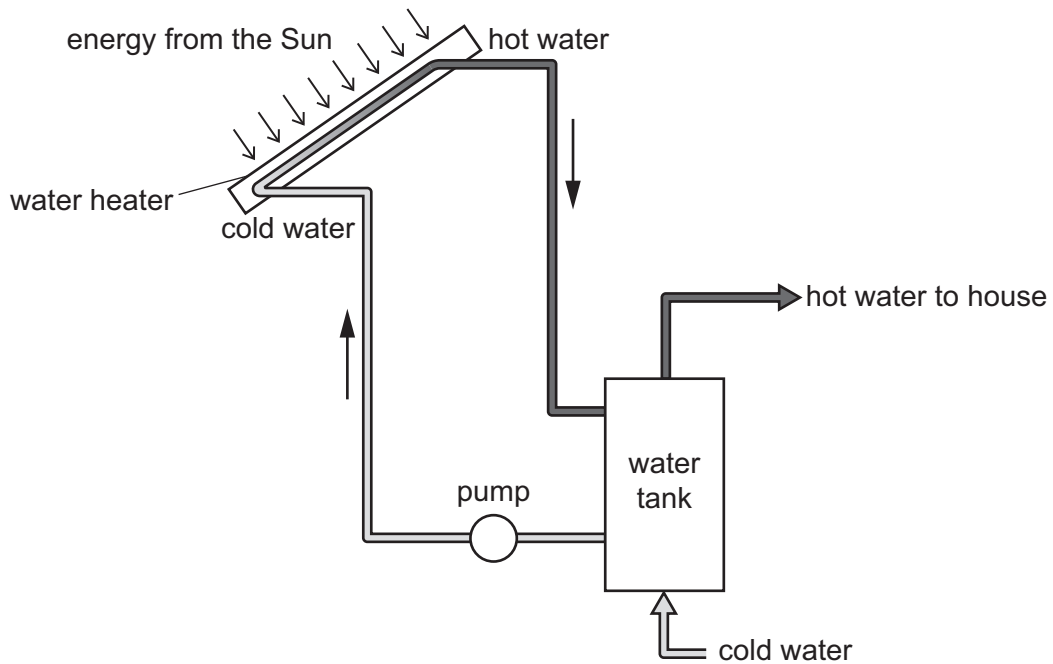
Explain how Aiko uses the equipment to model gas pressure **in a container**.

.....

.....

..... [2]

- 4 The diagram shows a water heating system that uses energy from the Sun.



- (a) (i) Write down **one** reason why this is an example of using a renewable energy resource.

.....
 [1]

- (ii) Suggest **one disadvantage** of using this method to heat water.

.....
 [1]

- (iii) Write down **one other renewable** energy resource.

..... [1]

- (iv) Write down **one non-renewable** energy resource.

..... [1]

(b) The picture shows a container made from plant material.



Which word describes this type of renewable material?

Circle the correct answer.

biochemical

bioplastic

cytoplasm

mineral

[1]

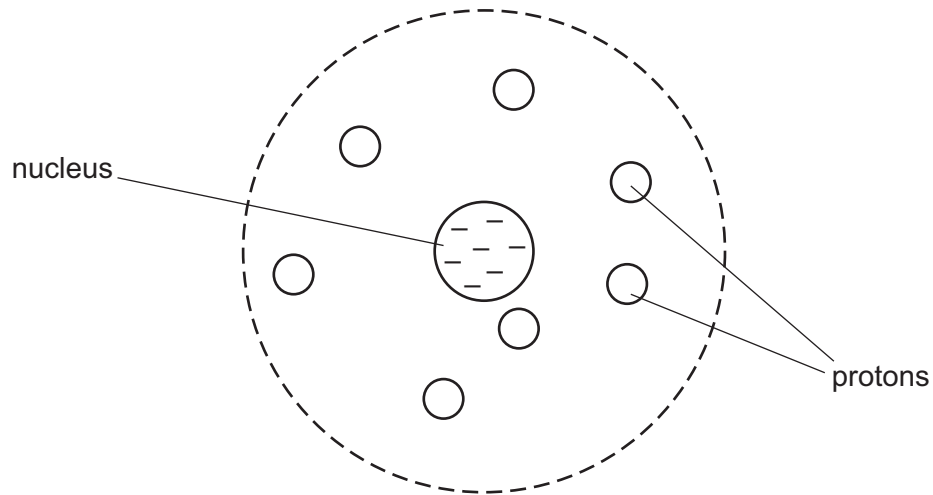
- 5 A balanced diet contains many different nutrients.

Draw a straight line from each **nutrient** to its correct **function in the body**.

nutrient	function in the body
calcium	for healthy eyesight
carbohydrate	to prevent dehydration
vitamin A	to make strong bones and teeth
protein	to provide energy and to be used as an energy store
water	for tissue growth and repair

[3]

6 Lily draws a diagram to show the model of an atom.



(a) (i) The diagram is **not** correct.

Write down **two** ways the diagram is **not** correct.

1

2

[2]

(ii) This model is named after a scientist.

Write down the name of this scientist.

..... [1]

(b) Explain how the particles in an atom are held together.

.....

.....

..... [2]

7 This question is about asteroids.

(a) The table shows some information about asteroids that have passed near to the Earth.

asteroid	date asteroid was nearest to the Earth	nearest distance from the Earth in km
A	February 2004	5481
B	March 2004	6542
C	October 2009	6259
D	October 2019	6242
E	August 2020	2946

(i) Which asteroid passed **nearer** than 5000 km from the Earth?

.....

[1]

(ii) Suggest why the distances in the table may **not** be accurate.

.....

.....

[1]

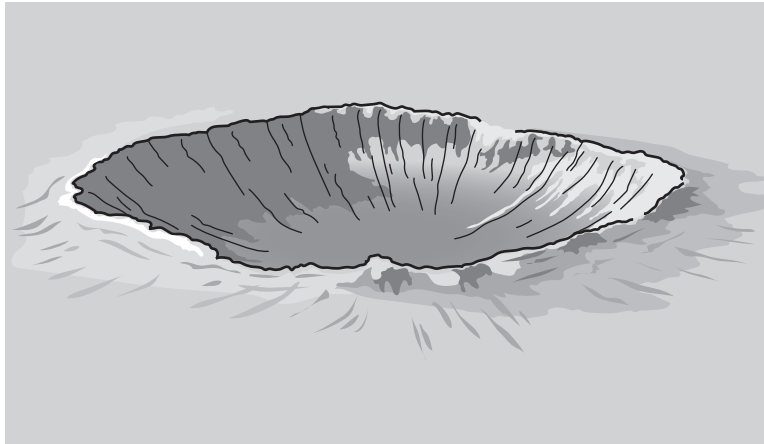
(iii) Describe how asteroids in space are formed.

.....

.....

[1]

- (b) The Moon has many craters formed when asteroids hit the surface of the Moon.

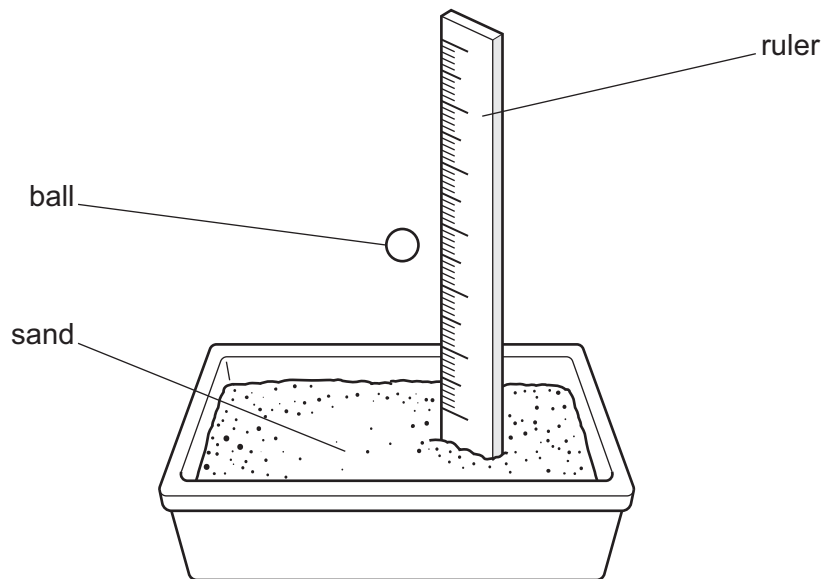


Ahmed makes this hypothesis about craters.

**‘The bigger the asteroid hitting the Moon,
the bigger the crater that is made.’**

Ahmed wants to test this hypothesis.

The diagram shows some of the equipment he uses.



Ahmed drops the ball and measures the size of the crater.

Describe how Ahmed uses this equipment to test his hypothesis.

.....

.....

..... [2]

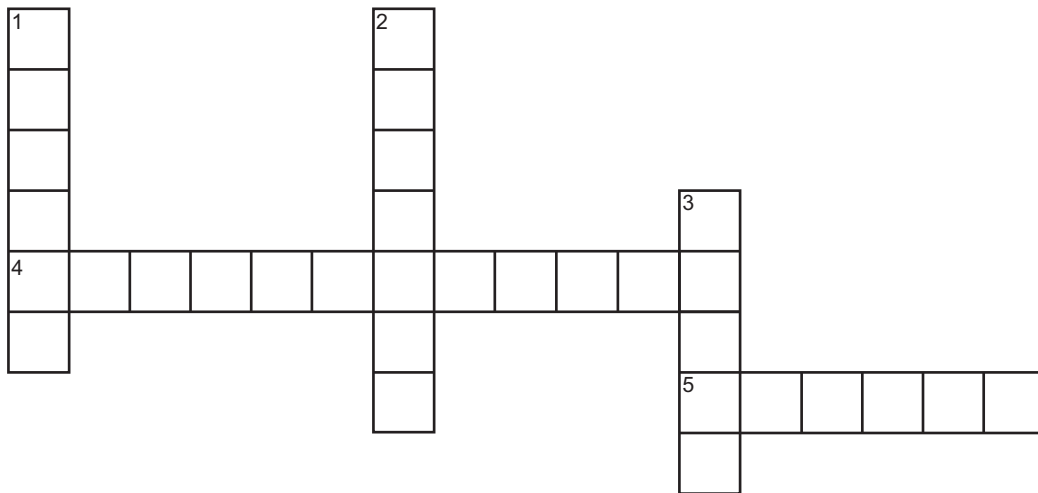
- 8 Safia makes an electromagnet.

Describe how Safia makes an electromagnet.

You may want to draw a diagram to help you answer the question.

[2]

9 Complete the crossword about aerobic respiration.



Clues

1 down

Which component of blood transports carbon dioxide?

.....

2 down

Which compound is a reactant of aerobic respiration?

.....

3 down

Which compound is a waste product of aerobic respiration?

.....

4 across

What is the name of the structure in a cell where aerobic respiration takes place?

.....

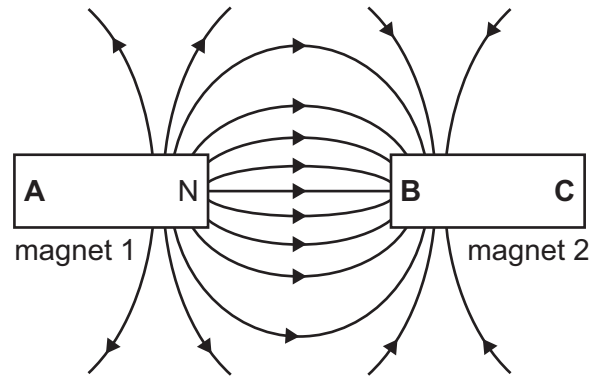
5 across

What is released during aerobic respiration?

.....

[4]

10 Blessy places two magnets next to each other as shown in the diagram.



The north pole, N, is labelled on magnet 1.

The other poles are labelled **A**, **B** and **C**.

(a) (i) What is shown by the pattern of lines around the two magnets?

..... [1]

(ii) Blessy writes some statements about the two magnets.

Tick (✓) the **correct** statements.

statement	
A is a N pole	
B is a S pole	
C is a N pole	

[1]

(b) Blessy brings pole **D** and pole **E** of two other magnets close to each other.

The magnets push each other away.

Which sentences are possible explanations?

Tick (✓) **two** boxes.

Pole **D** and pole **E** are both S poles.

☐

Pole **D** is a N pole and pole **E** is a S pole.

☐

Pole **D** is a S pole and pole **E** is a N pole.

☐

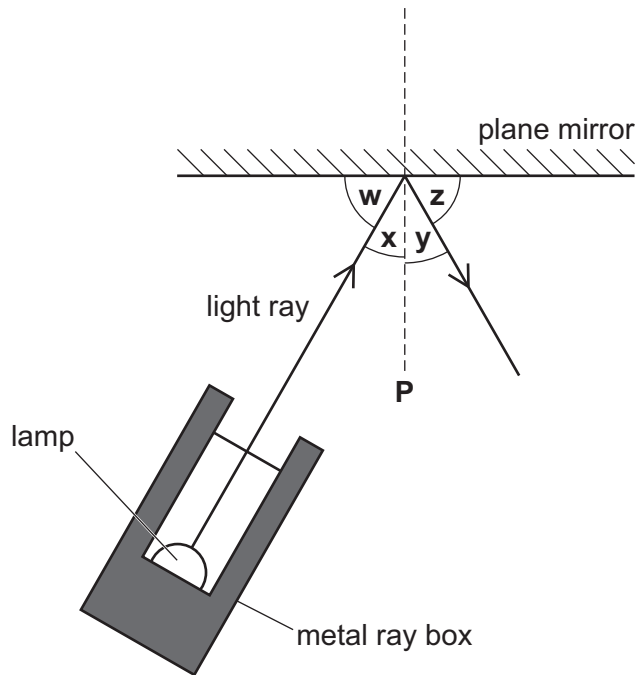
Pole **D** and pole **E** are both N poles.

☐

[2]

11 Mike investigates the reflection of a light ray at a plane mirror.

Look at his ray diagram.



(a) (i) Write down the name of the dotted line **P** in the diagram.

..... [1]

(ii) Which row in the table identifies the angle of incidence and the angle of reflection in Mike's diagram?

	angle of incidence	angle of reflection
A	w	x
B	w	y
C	x	y
D	x	z

row

[1]

- (b) Mike changes the angle of incidence and measures the angle of reflection.

The table shows the angles of incidence he uses.

Complete the table to predict the angles of reflection.

angle of incidence in °	angle of reflection in °
20	
30	
40	
50	

[1]

- (c) Mike uses a metal ray box connected to a power supply in his investigation.

Write down **one** safety hazard in Mike's investigation.

Describe how he avoids this hazard.

safety hazard

description of how Mike avoids the hazard

.....

[2]

12 Mia investigates solubility.

In her first experiment Mia:

- step 1 - pours 50 cm³ of water into a beaker
- step 2 - records the temperature of the water
- step 3 - adds 1g of a salt to the water
- step 4 - stirs the water until the salt dissolves
- step 5 - repeats steps 3 and 4 until no more salt dissolves
- step 6 - records the total mass of salt that dissolves.

Mia repeats the experiment four more times.

Each time she uses a different temperature of water.

Look at her results.

temperature of water in °C	mass of salt that dissolves in g
20	12
30	22
40	24
50	42
60	50

(a) What is the independent variable in this investigation?

..... [1]

(b) Describe the relationship between the temperature of the water and the mass of salt that dissolves.

.....
..... [1]

(c) Mia says,

‘I do not think my results are reliable.’

Write down **two** reasons why Mia is correct.

- 1
- 2

[2]



Science

Stage 8

Paper 2

2024

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INSTRUCTIONS

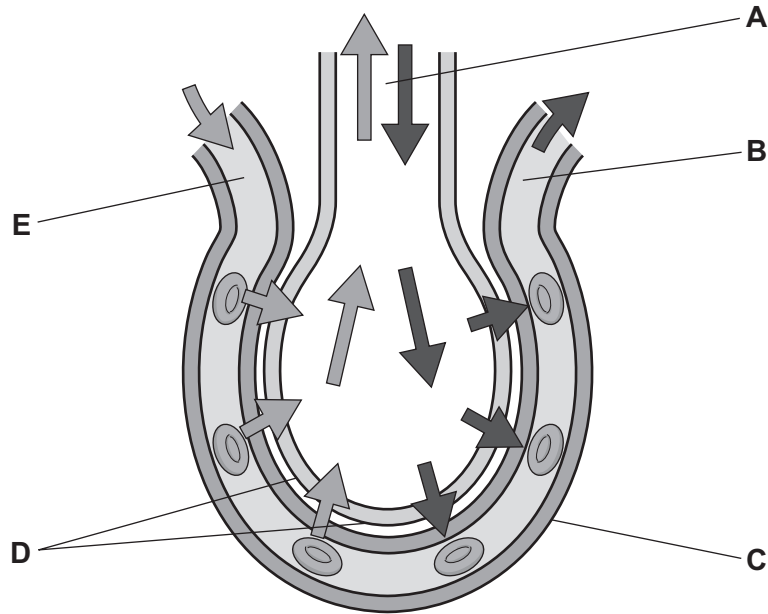
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1 Alveoli are air sacs found at the end of the bronchioles in the lungs.

(a) The diagram shows how gas is exchanged in alveoli.



Match the **description** to the correct **letter** shown in the diagram.

Write your answers in the table.

description	letter
place where the blood has the highest oxygen concentration	
blood capillary wall	
a surface where gas exchange happens	

[3]

(b) Alveoli are adapted to maximise gas exchange.

Describe **one adaptation** of the alveoli and explain how this helps to maximise gas exchange.

adaptation

.....

explanation

.....

[2]

(c) Red blood cells transport oxygen around the body.

Which mineral in the diet is needed to make red blood cells?

..... [1]

2 This question is about reactions of magnesium.

(a) Mia heats a piece of magnesium over a blue Bunsen flame.



(i) The magnesium reacts with a gas in the air.

Write down the name of this gas.

..... [1]

(ii) Describe one **safety risk** in this investigation and how to control the risk.

safety risk

how to control risk

..... [2]

(b) Magnesium reacts with dilute hydrochloric acid.

Circle **all** the products of this reaction.

carbon dioxide

chlorine

hydrogen

magnesium chloride

magnesium hydroxide

water

[2]

- 3 Safia uses an electromagnet to pick up steel pins.

She changes the current in the wire and records her results in the table.

current in A	number of pins
0.25	1
0.50	2
0.75	3
1.00	1
1.25	5

- (a) Circle the anomalous result in the table.

[1]

- (b) The current in the wire changes the strength of the electromagnet.

Suggest **one other** factor that changes the strength of the electromagnet.

..... [1]

- 4 Space has many parts.

Draw a straight line from each **part** to its correct **description**.

part	description
galaxy	clouds of particles in space
	object orbiting a planet
solar system	large group of stars
	small planets
stellar dust	planets orbiting around a star

[3]

5 Some industries produce toxic substances.

These toxic substances sometimes enter food chains.

Look at the food chain.

algae → mayfly → small fish → big fish → otter

(a) Suggest a habitat where this food chain is found.

..... [1]

(b) The table shows how the amount of toxic substance increases along the food chain.

organism	relative concentration of toxic substance
algae	1
mayfly	12
small fish	50
big fish	320
otter	5000

(i) Explain why the relative concentration of toxic substance is **much higher** in an otter than a big fish.

.....
 [1]

(ii) The amount of toxic substance increases along the food chain.

What is this process called?

..... [1]

(c) Write down **one impact** of increasing the concentration of toxic substances on an ecosystem.

.....
 [1]

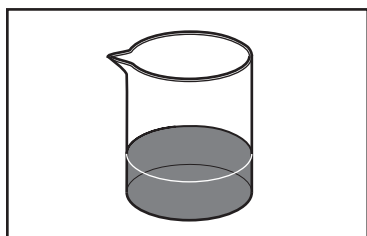
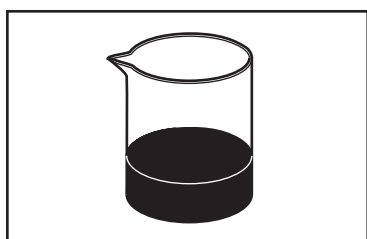
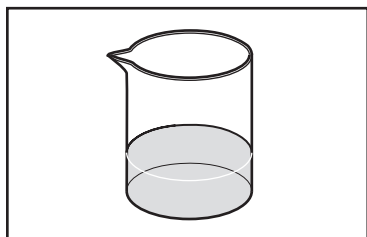
6 This question is about dissolving.

(a) When a blue food colour dissolves in water the solution turns blue.

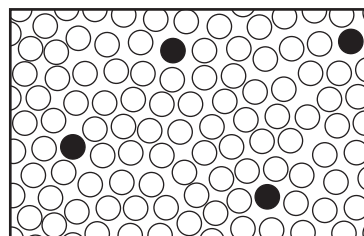
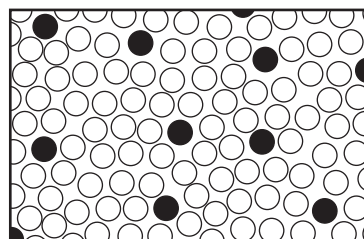
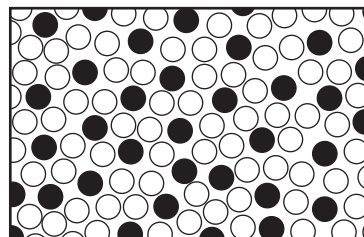
As the concentration of the blue food colour increases the solution becomes darker.

Draw a straight line from each **solution** to the correct **particle diagram** representing the solution.

solution



particle diagram



● = particle of blue food colour
○ = particle of water

[1]

(b) Circle the word in **bold** that makes each sentence correct.

Blue food colour dissolves in water.

The blue food colour is the **solute** / **solvent** / **soluble**.

Water is the **solute** / **solvent** / **soluble**.

[1]

7 Ahmed investigates primary and secondary colours.

(a) Which colour of light is produced when red light and green light are mixed?

Circle the correct answer.

yellow

blue

cyan

magenta

white

[1]

(b) Ahmed wears a magenta and green striped shirt.

Complete the sentences to explain how Ahmed's shirt looks in **blue light**.

In blue light, the magenta stripes look the colour

This is because magenta

.....

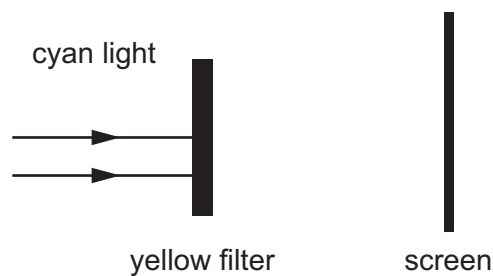
In blue light, the green stripes look the colour

This is because green

.....

[2]

(c) Ahmed passes cyan light through a yellow filter.



Explain which colour light is seen on the screen.

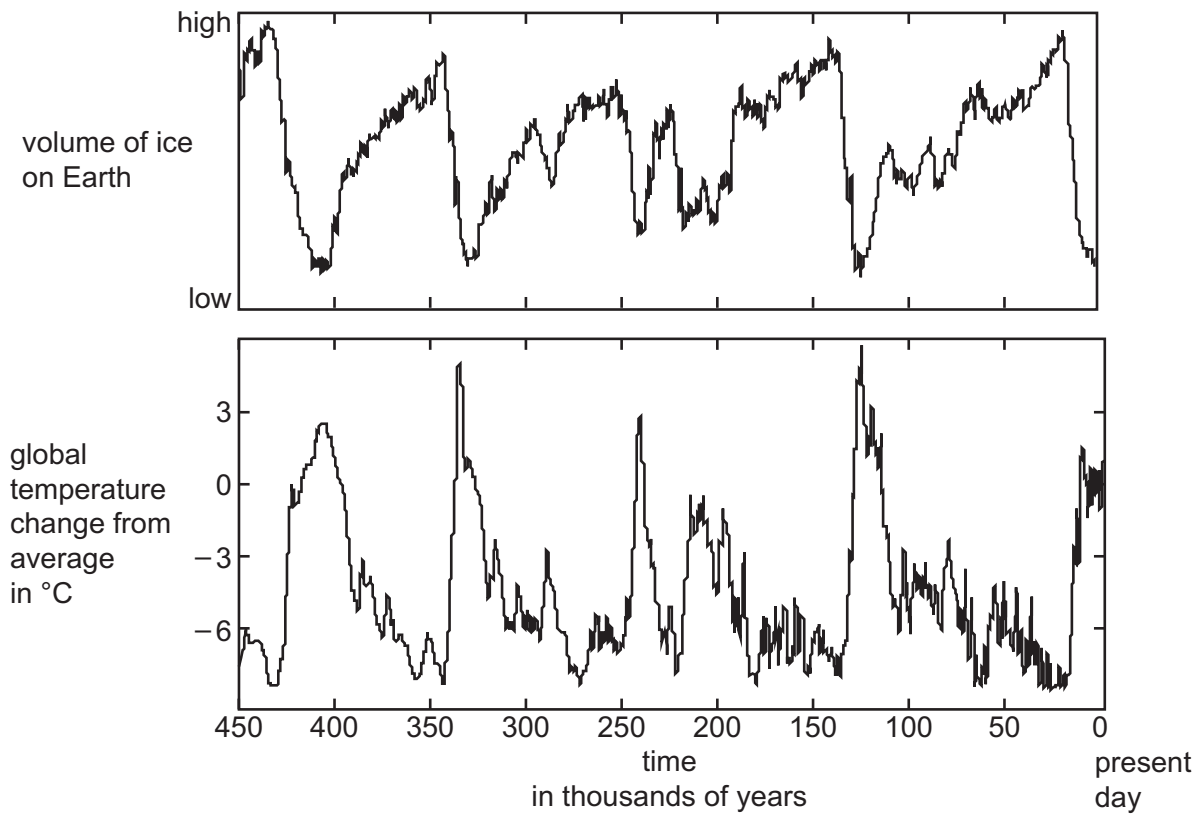
colour light

reason

.....

[1]

- 8 Look at the graphs about the volume of ice on Earth and global temperature change.



- (a) Use information from the graphs to answer these questions.

- (i) When was the **highest** global temperature?

..... thousand years ago [1]

- (ii) Describe the relationship between the global temperature and the volume of ice on Earth.

.....
 [1]

(b) Climate and weather are two different things.

Which statements are about **climate**?

Tick (✓) **all** the correct boxes.

average rainfall for a particular region	
changes in the atmosphere during a 24-hour period	
highest temperature recorded in a week	
long-term temperature pattern in a given area	
predicting when the next thunderstorm will happen	

[2]

9 Gold metal is used to make jewellery.

The purity of gold mixed with other metals is measured using a system called the carat rating.

The table shows the percentage of gold and the percentage of other metals in different carat ratings.

carat rating	minimum percentage of gold in the metal	maximum percentage of other metals mixed with the gold
24	100.0	0.0
22	91.6	8.4
18	75.0	25.0
14	58.5	41.5
10	41.7	58.3

A piece of gold mixed with other metals has a mass of 200 g.

The mass of gold in this mixture is 86.5 g.

(a) Calculate the percentage of gold in the mixture.

percentage of gold % [1]

(b) What is the carat rating for this mixture of gold and other metals?

carat rating [1]

10 Complete the sentences about diffusion.

Diffusion occurs in gases and

When perfume is sprayed in a room, the perfume particles and air particles move and

.....

Diffusion occurs more quickly if a gas is hotter because the gas particles move

.....

[2]

11 Yuri investigates the water content of potatoes.

Yuri:

step 1 - cuts a piece of potato with a sharp knife into a cube of mass 10.0 g

step 2 - heats the cube of potato in a microwave for 1 minute

step 3 - uses a balance to find the mass of the cube of potato

step 4 - repeats steps 2 and 3 until the cube of potato has been heated for a total of 10 minutes.

(a) Here are his results.

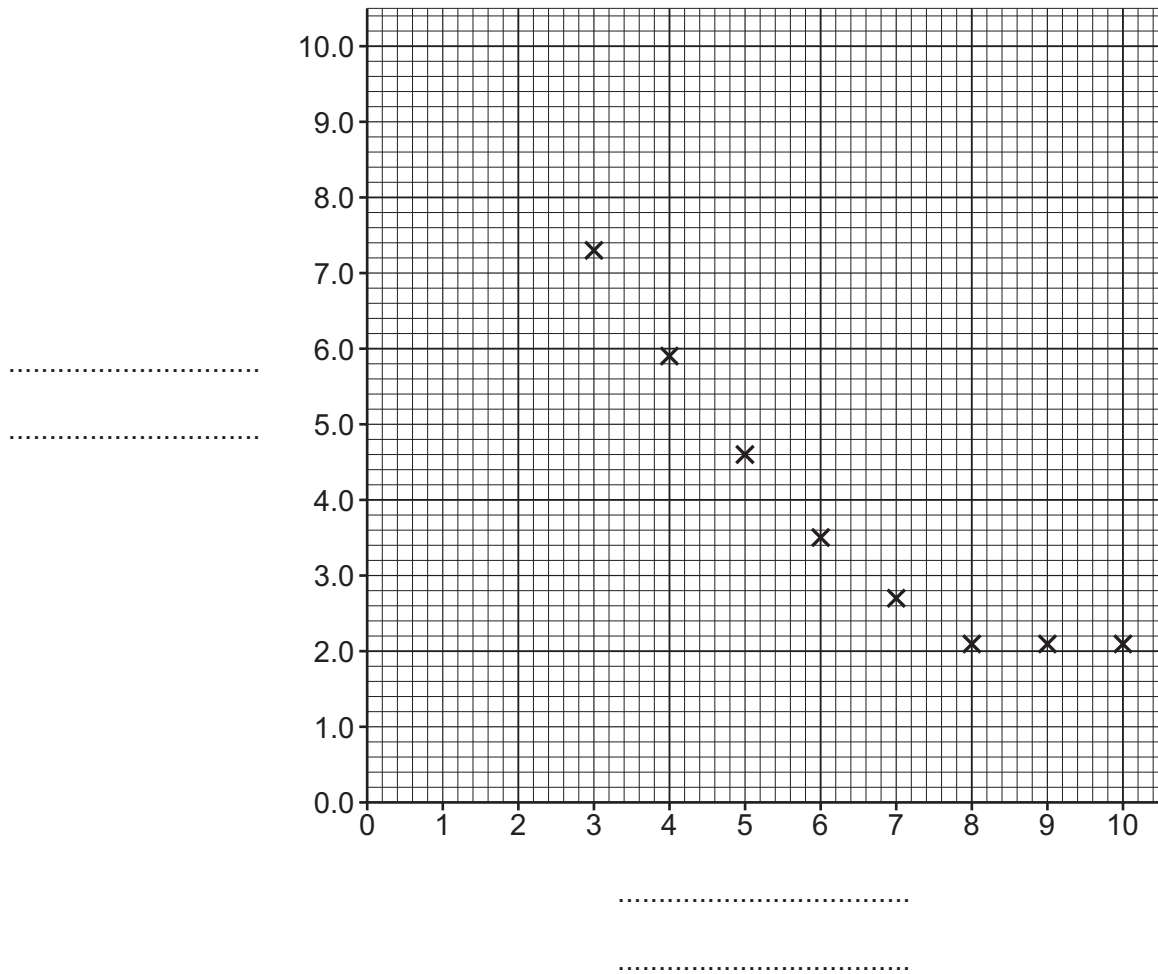
time in minutes	mass of cube of potato in g
0	10.0
1	9.4
2	8.5
3	7.3
4	5.9
5	4.6
6	3.5
7	2.7
8	2.1
9	2.1
10	2.1

Look at the graph of his results.

The last eight masses from the table have been plotted on the graph.

Complete the graph by:

- labelling the axes
- plotting the first three masses from the table on the graph
- drawing a curve of best fit.



[4]

(b) Yuri says,

**'I have read that vegetables are 90% water.
I predict 90% of this potato will be water.'**

Tick (✓) to show if the results support Yuri's prediction.

yes

☐

no

☐

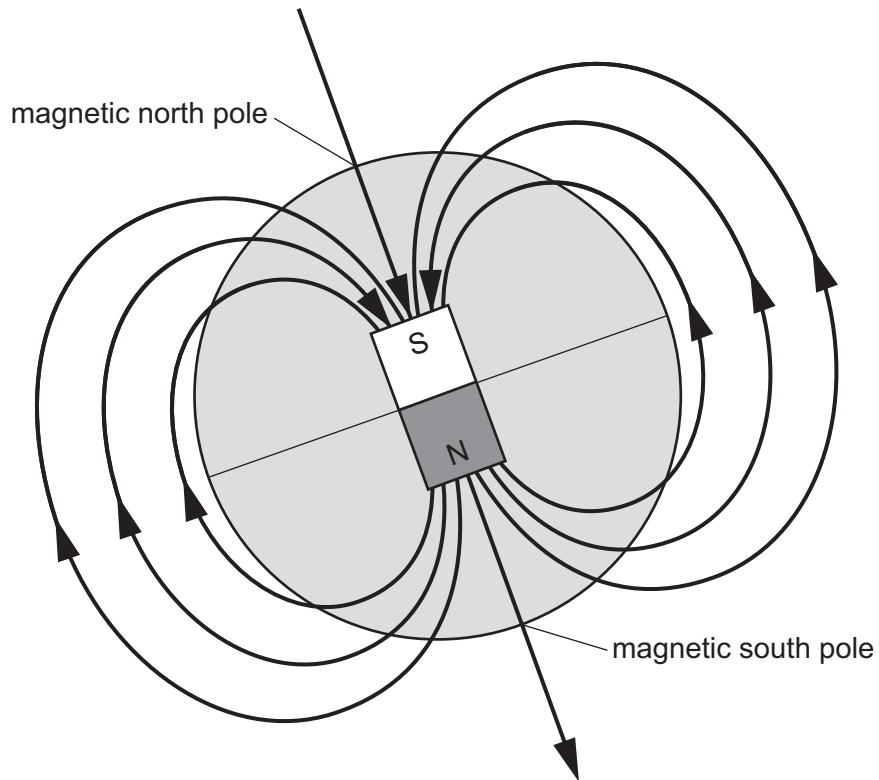
Explain your answer.

.....

.....

[1]

12 The diagram shows an analogy to describe the Earth's magnetism.



(a) Which part of the Earth's structure is represented by the bar magnet?

..... [1]

(b) What do the arrows represent?

..... [1]

13 Mike investigates reactions.

In his first experiment Mike:

- pours some water into a beaker
- measures the temperature of the water
- adds 1g of a solid to the water
- stirs the mixture until the temperature stops changing.

Mike repeats the experiment with four other solids.

Look at his results.

solid	temperature at the start in °C	temperature at the end in °C
A	20	20
B	21	28
C	19	15
D	18	19
E	21	27

(a) (i) How many solids have an endothermic reaction with water?

..... [1]

(ii) Mike has **not** done a fair test.

Suggest **one** way Mike could improve his investigation to make it a fair test.

.....
 [1]

(iii) Mike describes solid **A** as being inert.

What is meant by the term inert?

..... [1]

(b) One example of an exothermic reaction is adding calcium oxide to water.

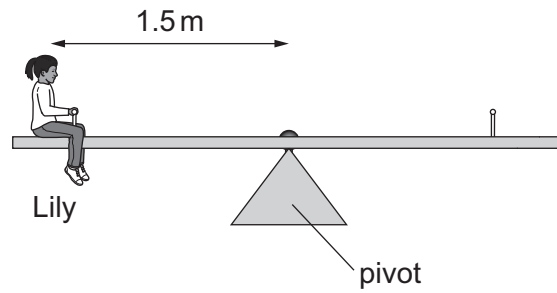
The product is calcium hydroxide.

Write the word equation for this reaction.

..... [1]

14 Lily and Blessy play on a seesaw.

(a) Lily sits at a distance of 1.5 m from the pivot.



Lily weighs 500 N.

Calculate the moment of Lily's weight about the pivot.

Include the unit in your answer.

moment of Lily's weight = unit [3]

(b) Blessy sits on the seesaw with Lily.

Blessy weighs 600 N.

The seesaw is balanced.

Calculate the distance Blessy sits from the pivot.

Use ideas about the principle of moments.

distance from pivot m [2]